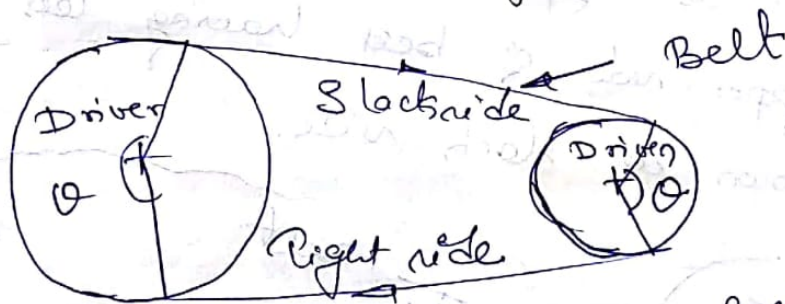


Module-4

* Belt drives :-

- Belt drives are used to transmit power or motion from one shaft to the other by means of a thin inextensible belt running over two pulleys.
- The arrangement consists of two pulleys mounted on two different shafts.
- One shaft called the driving shaft receives power from the mains and transmits it to another shaft called driven shaft.
- The pulley mounted on the driving shaft is called driving pulley or driver, while the other pulley mounted on a shaft to which the power is to be transmitted is called the driven pulley or follower.
- The belt passing over the two pulleys is kept in tension so as to avoid slip over the pulleys.



- Tight side:- The side of the belt where there is maximum tension is called tight side. (T_1)
- Slack side:- The side of the belt where there is minimum tension
- Arc / Angle of contact:- It is the portion of belt which is in contact with the pulley surface (θ).

* Classification of Belt drives :-

There are mainly two types of belt drives namely, 1) open belt drive

2) Cross belt drive.

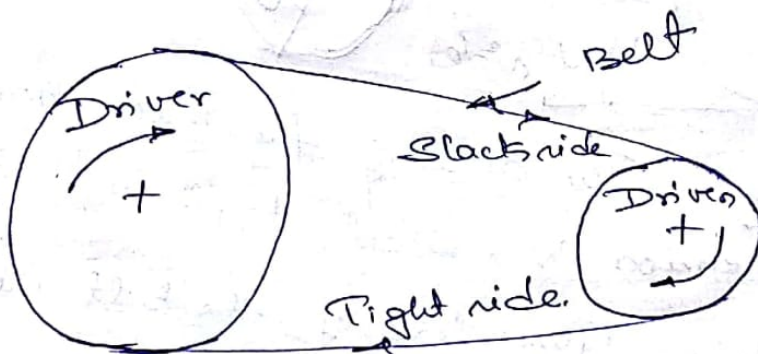
→ Open Belt drive :-

→ The arrangement of open belt drive is shown in fig. Both driver and driven pulleys rotate in the same direction. Belt passes over driver & driven pulleys.

→ The driver pulley pulls the belt from one side and delivers it to the other side. For this reason, the tension in the lower side of the belt is more than that in the upper side.

→ The lower side of belt having more tension is known as tight side.

→ The upper side of belt having less tension is known as slack side.

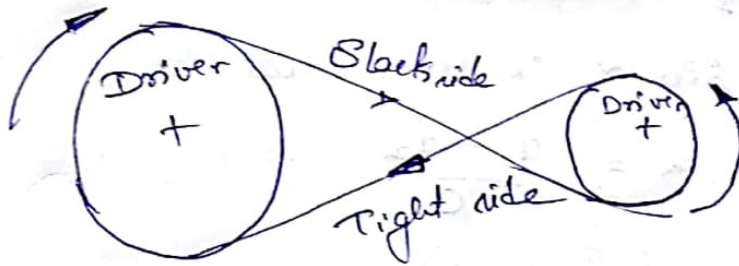


→ Cross Belt drive :-

The arrangement of cross belt drive is shown in figure. Driven rotates in opposite

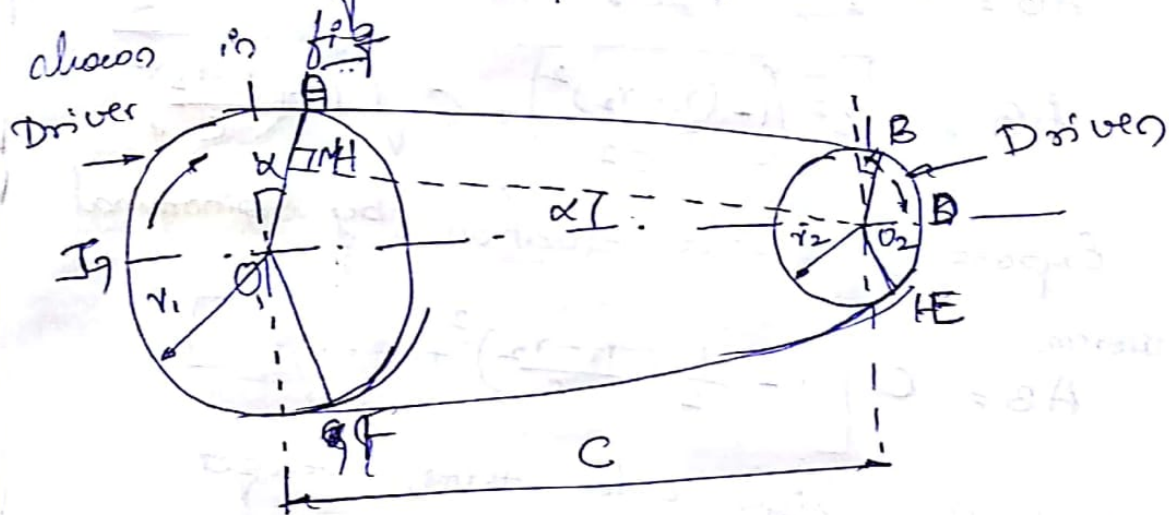
same direction to that of driver.

→ The upper side is slack side & the lower side is tight side in the arrangement.



* Length of an Open Belt Drive

The arrangement of an open belt drive as



Let r_1 = radius of larger pulley

r_2 = radius of smaller pulley

C = Center Distance b/w Driver & Driven Pulley

L = Total length of belt

α = Angle subtended by each common tangent with O_1, O_2 .

∴ Total length of belt $L = \text{Arc EGA} + AB + \text{Arc BDE} + EF$

Total length of belt $L = \text{Arc EGA} + AB + \text{Arc BDE} + EF$
 $= 2 (\text{Arc GA} + AB + \text{Arc BD})$

Draw O_2H parallel to AB so that $O_1H = r_1 - r_2$ from the figure

$$\sin \alpha = \frac{O_1H}{O_1O_2} = \frac{O_1A - HA}{O_1O_2} = \frac{r_1 - r_2}{C}$$

Since $\sin \alpha$ is very small $\therefore \alpha$

$$\sin \alpha = \alpha = \frac{r_1 - r_2}{C}$$

$$\text{Arc } QA = r_1 \left(\frac{\pi}{2} + \alpha \right)$$

$$\text{Arc } BD = r_2 \left(\frac{\pi}{2} - \alpha \right)$$

$$AB = HO_2 = \sqrt{O_1O_2^2 - O_1H^2} = \sqrt{C^2 - (r_1 - r_2)^2}$$

$$AB = \sqrt{C^2 \left(1 - \frac{(r_1 - r_2)^2}{C^2} \right)} = C \sqrt{1 - \left(\frac{r_1 - r_2}{C} \right)^2}$$

Expanding above equation by binomial theorem.

$$AB = C \left[1 - \frac{1}{2} \left(\frac{r_1 - r_2}{C} \right)^2 + \dots \right]$$

Neglecting high order terms, we get

$$AB = C - \frac{1}{2} \left[\frac{(r_1 - r_2)^2}{C} \right]$$

$\therefore$ Total length of open belt driven

$$L = 2 \left[r_1 \left(\frac{\pi}{2} + \alpha \right) + C - \frac{1}{2} \frac{(r_1 - r_2)^2}{C} + r_2 \left(\frac{\pi}{2} - \alpha \right) \right]$$

$$= 2 \left[r_1 \frac{\pi}{2} + r_1 \alpha + C - \frac{(r_1 - r_2)^2}{2C} + r_2 \frac{\pi}{2} - r_2 \alpha \right]$$

$$= 2 \left[\frac{\pi}{2} (r_1 + r_2) + \alpha (r_1 - r_2) + C - \frac{(r_1 - r_2)^2}{2C} \right]$$

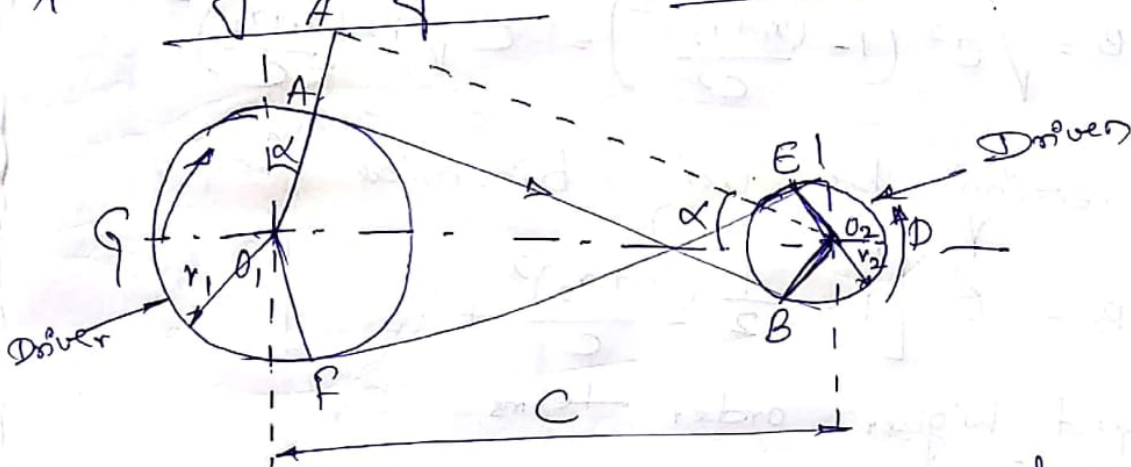
$$L = \pi(r_1 + r_2) + 2\alpha(r_1 - r_2) + 2C - \frac{(r_1 - r_2)^2}{C}$$

$$= \pi(r_1 + r_2) + 2 \times \frac{(r_1 - r_2)}{C} (r_1 - r_2) + 2C - \frac{(r_1 - r_2)^2}{C}$$

$$= \pi(r_1 + r_2) + 2 \frac{(r_1 - r_2)^2}{C} + 2C - \frac{(r_1 - r_2)^2}{C}$$

$$L = \pi(r_1 + r_2) + 2C + \frac{(r_1 - r_2)^2}{C}$$

* Length of Crossed belt drive



Consider a Crossed-belt drive as shown in fig

Let r_1 - Radius of Driver pulley

r_2 - Radius of Driven pulley

C - Center distance b/w two pulleys

α - Angle subtended by each common tangent with O_1, O_2

Draw O_2H parallel to AB so that $O_1H = r_1 + r_2$

Total length of the belt

$$L = \text{arc } FGA + AB + \text{Arc } BDE + EF$$

$$= 2(\text{Arc } GA + AB + \text{Arc } BD)$$

from fig $\sin \alpha = \frac{O_1 H}{O_2 O_1} = \frac{O_1 A + HA}{O_1 O_2} = \frac{r_1 + r_2}{C}$

Since α is very small

$$\sin \alpha = \alpha = \frac{r_1 + r_2}{C}$$

$$\text{Arc GA} = r_1 \left(\frac{\pi}{2} + \alpha \right)$$

$$\text{Arc BD} = r_2 \left(\frac{\pi}{2} + \alpha \right)$$

$$AB = \sqrt{O_1 O_2^2 - O_1 H^2} = \sqrt{C^2 - (r_1 + r_2)^2}$$

$$AB = \sqrt{C^2 \left(1 - \frac{(r_1 + r_2)^2}{C^2} \right)} = C \sqrt{1 - \left(\frac{r_1 + r_2}{C} \right)^2}$$

Expressing by using binomial theorem

$$AB = C \left[1 - \frac{1}{2} \frac{(r_1 + r_2)^2}{C^2} + \dots \right]$$

Neglect higher order terms

$$AB = C - \frac{(r_1 + r_2)^2}{2C}$$

$\therefore$ Total length of belt:

$$L = 2 \left[r_1 \left(\frac{\pi}{2} + \alpha \right) + C - \frac{(r_1 + r_2)^2}{2C} + r_2 \left(\frac{\pi}{2} + \alpha \right) \right]$$

$$= 2 \left[r_1 \frac{\pi}{2} + r_1 \alpha + C - \frac{(r_1 + r_2)^2}{2C} + r_2 \frac{\pi}{2} + r_2 \alpha \right]$$

$$= 2 \left[\frac{\pi}{2} (r_1 + r_2) + \alpha (r_1 + r_2) + C - \frac{(r_1 + r_2)^2}{2C} \right]$$

$$= \pi (r_1 + r_2) + 2\alpha (r_1 + r_2) + 2C - \frac{(r_1 + r_2)^2}{C}$$

But $\alpha = \frac{r_1 + r_2}{C}$

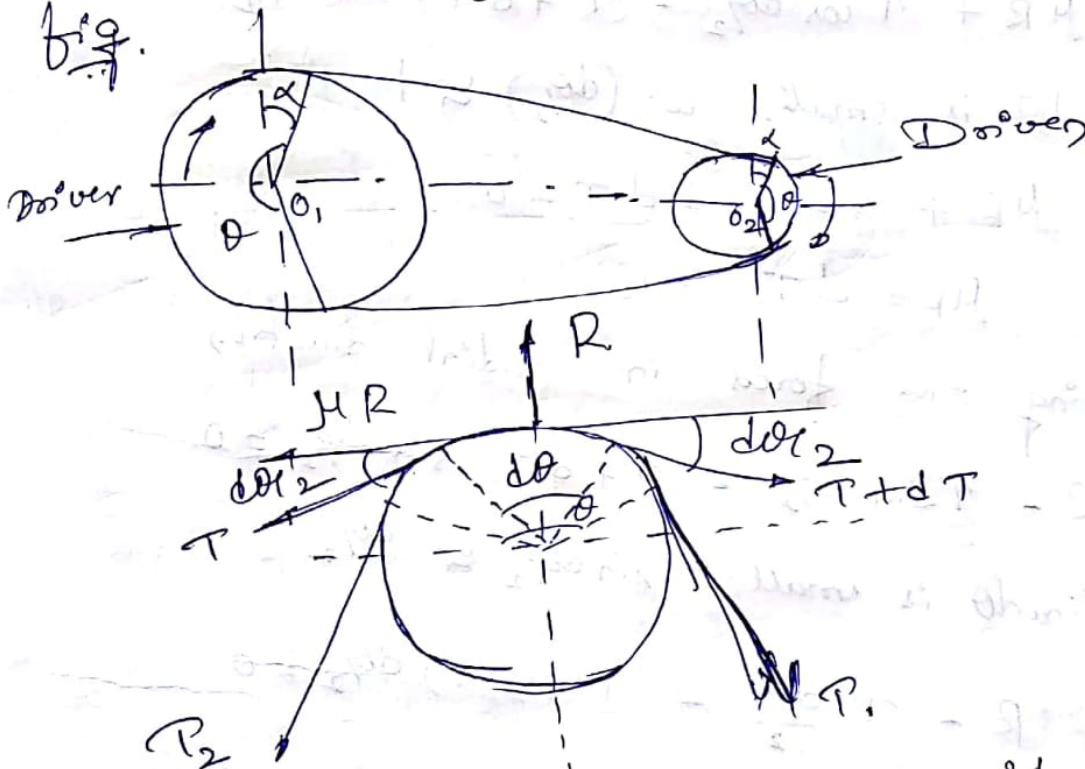
$$\therefore L = \pi(r_1 + r_2) + 2 \frac{(r_1 + r_2)}{C} (r_1 + r_2) + 2C - \frac{(r_1 + r_2)^2}{C}$$

$$= \pi(r_1 + r_2) + 2 \frac{(r_1 + r_2)^2}{C} + 2C - \frac{(r_1 + r_2)^2}{C}$$

$$L = \pi(r_1 + r_2) + 2C - \frac{(r_1 + r_2)^2}{C}$$

* Ratio of belt tensions :-

Consider a open flat belt drive as shown in fig.



Let T_1 = Tension on tight side

T_2 = Tension on slack side

θ = Angle of lap of the belt over the pulley.

μ = Co-efficient of friction b/w belt and pulley.

Consider a short length of belt subtending an angle $d\theta$ at the centre of the pulley.

Let R = Normal reaction b/w the element length of belt & the pulley

T = Tension on slack side of the element

$T + dT$ = Tension on tight side of the element

dT = Increase in tension on tight side.

Resolving the forces in tangential direction

$$\mu R + T \cos \frac{d\theta}{2} - (T + dT) \cos \frac{d\theta}{2} = 0$$

As $d\theta$ is small, $\cos(\frac{d\theta}{2}) \approx 1$

$$\mu R + T - T - dT = 0$$

$$\mu R = dT \quad \text{--- (1)}$$

Resolving the forces in radial direction

$$R - T \sin \frac{d\theta}{2} - (T + dT) \sin \frac{d\theta}{2} = 0$$

Since $d\theta$ is small, $\sin \frac{d\theta}{2} \approx \frac{d\theta}{2}$

$$R - T \frac{d\theta}{2} - (T + dT) \frac{d\theta}{2} = 0$$

$$R - T \frac{d\theta}{2} - T \frac{d\theta}{2} - dT \frac{d\theta}{2} = 0$$

[$dT d\theta$ is very small therefore, it is neglected]

~~$$R - T d\theta = 0$$~~

$$R = T d\theta \quad \text{--- (2)}$$

Substituting equation (2) in Eqn (1)

$$\mu \times T d\theta = dT$$

$$\frac{dT}{T} = \mu d\theta$$

Integrating b/w. T_2 , T_1 & 0 , θ

$$\int_{T_2}^{T_1} \frac{dT}{T} = \mu \int_0^\theta d\theta$$

$$\ln T_1 - \ln T_2 = \mu \theta$$

$$\ln \left(\frac{T_1}{T_2} \right) = \mu \theta$$

$$\boxed{\frac{T_1}{T_2} = e^{\mu \theta}}$$

* Velocity Ratio

It is defined as the ratio of the speed of the driven pulley to the speed of the driving pulley.

$$\therefore \text{Velocity ratio} = VR = \frac{\text{Speed of driven pulley}}{\text{Speed of driver pulley}}$$

$$VR = \frac{N_2}{N_1}$$

we have
Linear speed of belt = Circumferential speed of driving pulley = Circumferential speed of driven pulley

$$= \pi d_1 N_1 = \pi d_2 N_2$$

$$\frac{N_2}{N_1} = \frac{d_1}{d_2}$$

$$\boxed{VR = \frac{N_2}{N_1} = \frac{d_1}{d_2}}$$

If belt thickness is considered, then

$$VR = \frac{d_1 + t}{d_2 + t} = \frac{N_2}{N_1}$$

If slip occur

$$VR = \frac{N_2}{N_1} = \frac{d_1}{d_2} \left[1 - \frac{s}{100} \right]$$

$$\text{or } VR = \frac{N_2}{N_1} = \frac{d_1 + t}{d_2 + t} \left[1 - \frac{s}{100} \right]$$

* Power transmitted by the belt

$$P = \text{Net tension} \times \text{distance moved/s}$$

$$P = (T_1 - T_2) \times V \text{ in watt}$$

* Advantages of V-belt

1. Compact & give high velocity ratio
2. Positive & reliable drive
3. Provides shock absorption b/w driver & driven shaft
4. There is no joint problems as the drive is of endless.
5. Because of wedging action in the grooves loss of power due to slip is less.

* Disadvantages of V-belts

1. Initial cost is more
2. Cannot be used when the centre distance is large
3. Improper belt tensioning.

Problems.

1) It is required to drive a shaft A at 600 rpm by a belt using a pulley of 150 mm diameter on another parallel shaft B running at 240 rpm. Determine the diameter of the pulley on the shaft A? Also find the velocity ratio.

Given data :-

Driving shaft (B)

$$N_1 = 240 \text{ rpm}$$

$$d_1 = 150 \text{ mm}$$

Driven shaft (A)

$$N_2 = 600 \text{ rpm}$$

$$d_2 = ?$$

a) W.K.T. for the belt drive,

$$\text{Velocity ratio} = \frac{N_2}{N_1} = \frac{d_1}{d_2}$$

$$d_2 = d_1 \times \frac{N_1}{N_2} = 150 \times \frac{240}{600} = 60 \text{ mm}$$

$$\text{Velocity ratio} = \frac{N_2}{N_1} = \frac{600}{240} = 2.5$$

$$\therefore \text{Velocity ratio} = 2.5 : 1$$

2) The sum of diameters of two pulleys is 1000 mm & the pulleys are connected by belt. If the pulleys rotate at 600 rpm & 1800 rpm. determine the diameters of each pulley.

Given data :-

$$d_1 + d_2 = 1000 \text{ mm}, \quad N_1 = 600 \text{ rpm}, \quad N_2 = 1800 \text{ rpm}$$

$$\therefore \text{Velocity ratio} = \frac{N_2}{N_1} = \frac{d_1}{d_2}$$

$$\frac{1800}{600} = \frac{d_1}{d_2} \Rightarrow d_1 = 3d_2$$

But $d_1 + d_2 = 1000$

$\therefore 3d_2 + d_2 = 1000$

$$d_2 = 250 \text{ mm}$$

$d_1 = 3d_2 = 750 \text{ mm}$

③ Two pulleys are connected by a crossed belt. The velocity ratio of the drive is 3. The driving pulley runs at 1000 rpm & 120 cm diameter. Find the speed and the diameter of the driven pulley. Draw the line diagram of the drive. Given data ..

Velocity ratio = $\frac{N_2}{N_1} = 3$

$N_1 = 1000 \text{ rpm}$, $d_1 = 120 \text{ cm} = 1200 \text{ mm}$

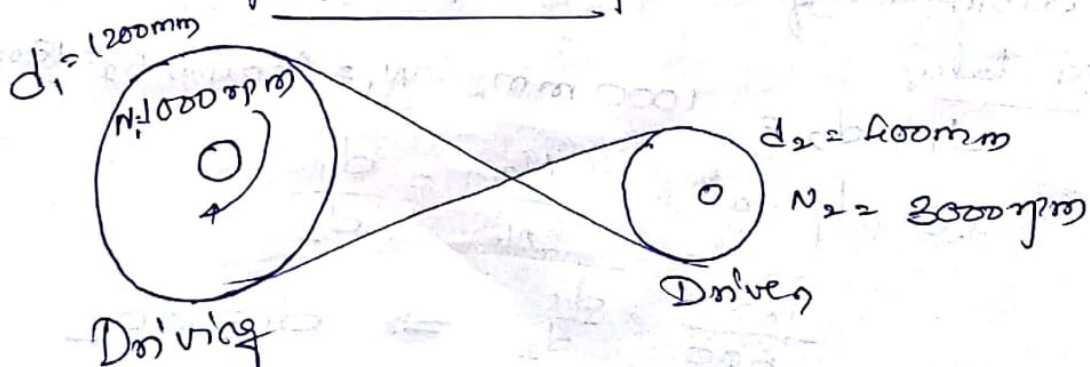
$\therefore \text{Velocity ratio} = \frac{N_2}{N_1} = 3 = \frac{1000}{N_2} \times \frac{N_2}{1000}$

$N_2 = 3000 \text{ rpm}$

$\frac{N_2}{N_1} = \frac{d_1}{d_2}$

$d_2 = d_1 \times \frac{N_1}{N_2} = 1200 \times \frac{1000}{3000}$

$$d_2 = 400 \text{ mm}$$



- ④ In a belt driven system of power transmission the driving pulley running at 500rpm drives another pulley 30cm in diameter at 760rpm. Find the diameter of the driving pulley.

Given data :-

Driving Pulley

$$N_1 = 500 \text{ rpm}$$

$$d_1 = ?$$

Driven Pulley

$$N_2 = 760 \text{ rpm}$$

$$d_2 = 30 \text{ cm} = 300 \text{ mm}$$

$$\text{Velocity ratio} = \frac{N_2}{N_1} = \frac{d_1}{d_2}$$

$$\frac{760}{500} = \frac{d_1}{300}$$

$$\boxed{d_1 = 456 \text{ mm}}$$

- ⑤ A prime mover running at 2400rpm drives a DC generator by a belt drive. The diameter of the pulley on the output shaft of the prime mover is 1600mm & that of the generator shaft is 800mm. Determine the speed of the generator shaft in the following

Case 1) Neglecting thickness of belt

2) Considering belt thickness as 6mm.

3) Considering thickness of belt & a slip of 3%.

4) Velocity of belt considering belt thickness.

Given data :-

Driver

$$N_1 = 2400 \text{ rpm}$$

$$d_1 = 1600 \text{ mm}$$

Driven

$$N_2 = ?$$

$$d_2 = 800 \text{ mm}$$

a) Neglecting belt thickness

$$\frac{N_2}{N_1} = \frac{d_1}{d_2} \Rightarrow N_2 = N_1 \frac{d_1}{d_2}$$

$$N_2 = 240 \times \frac{1600}{800} \quad \boxed{N_2 = 480 \text{ rpm}}$$

b) Considering belt thickness, $t = 6 \text{ mm}$

$$\frac{N_2}{N_1} = \frac{d_1 + t}{d_2 + t} \Rightarrow N_2 = N_1 \times \frac{(d_1 + t)}{(d_2 + t)}$$

$$N_2 = 240 \times \left(\frac{1600 + 6}{800 + 6} \right) = \boxed{N_2 = 478.2 \text{ rpm}}$$

c) Considering belt thickness & slip of 3%

$$\frac{N_2}{N_1} = \frac{d_1 + t}{d_2 + t} \left(\frac{100 - S}{100} \right)$$

$$N_2 = 240 \times \left(\frac{1600 + 6}{800 + 6} \right) \left(\frac{100 - 3}{100} \right)$$

$$\boxed{N_2 = 463.8 \text{ rpm}}$$

d) Velocity of belt considering thickness

$$V = \frac{\pi (d_1 + t) N_1}{60} = \frac{\pi (1600 + 6) \times 10^3 \times 240}{60}$$

$$\boxed{V = 20.86 \text{ m/s}}$$

e) In a cross belt drive the difference in tension between tight and slack sides is 1200 N . If the angle of contact is 160° & the coefficient of friction is 0.28 . Find the tension between slack and tight sides.

Given data

$$T_1 - T_2 = 1200 \text{ N}$$

$$\theta = 160^\circ = \frac{160 \times \pi}{180} = 2.792 \text{ radians}$$

$$\mu = 0.28$$

W.K.T $\frac{T_1}{T_2} = e^{\mu \theta}$

$$\frac{T_1}{T_2} = e^{(0.28 \times 2.792)}$$

$$T_1 = 2.185 T_2$$

$$T_1 = 1200 + T_2$$

$$2.185 T_2 = 1200 + T_2$$

$$T_2 = 1012.65 \text{ N}$$

Slack side tension

$$T_1 = 1200 + 1012.65$$

$$T_1 = 2212.65 \text{ N}$$

Tight side tension

④ In a belt drive the ratio of tensions is 2 and the slack side tension is 500 N. If the speed and diameter of the driven pulley are 200 rpm and 120 cm respectively find the power transmitted.

Given data:

$$\frac{T_1}{T_2} = 2, T_2 = 500 \text{ N}, N_2 = 200 \text{ rpm}$$

$$d_2 = 120 \text{ cm} = 1.2 \text{ m}$$

Power Transmitted (P)

$$P = \frac{(T_1 - T_2) V}{1000} \text{ kW}$$

$$V = \pi d_1 N_1 = \pi d_2 N_2$$

$$V = \pi \times 200 \times 1.2$$

$$V = 753.98 \text{ m/min}$$

$$P = \frac{(1000 - 500)}{1000} \times 12.56$$

$$\boxed{P = 6.28 \text{ kW}}$$

⑧ Two pulleys of 50cm and 25 cm diameter are to be coupled by a belt drive. If the distance b/w their axis is 1.5m, find the length of the belt for a) open & b) cross drives.

Given data: $d_1 = 50 \text{ cm} = 0.5 \text{ m}$, $r_1 = 0.25 \text{ m}$
 $d_2 = 25 \text{ cm} = 0.25 \text{ m}$, $r_2 = 0.125 \text{ m}$
 $C = 1.5 \text{ m}$

length of belt for open belt drive

$$L = 2C + \pi(r_1 + r_2) + \frac{(r_1 - r_2)^2}{C}$$

$$= 2(1.5) + \pi(0.25 + 0.125) + \frac{(0.25 - 0.125)^2}{1.5}$$

$$\boxed{L = 4.188 \text{ m}}$$

length of belt for cross belt drive

$$L = 2C + \pi(r_1 + r_2) + \frac{(r_1 + r_2)^2}{C}$$

$$= 2(1.5) + \pi(0.25 + 0.125) + \frac{(0.25 + 0.125)^2}{1.5}$$

$$\boxed{L = 4.271 \text{ m}}$$

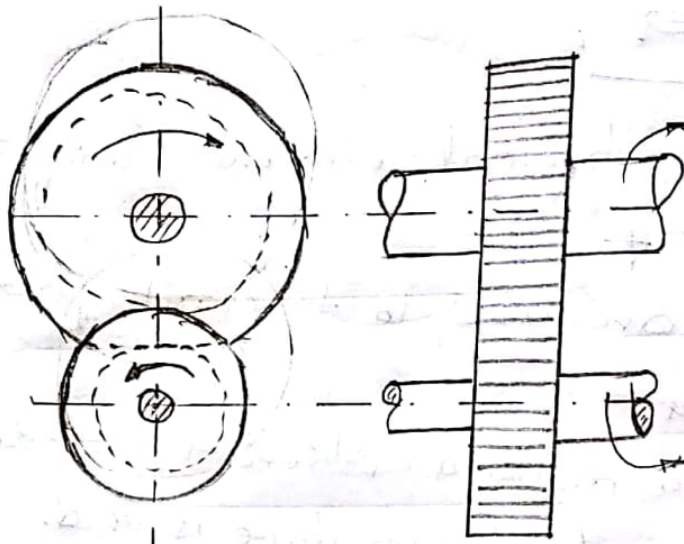
Gear Drives :-

- Gear is a mechanical element having teeth on its periphery
- Gear drives are used to transmit power over short distances.
- In precision machines, where the constant velocity ratio is of important gear drive is used.
- Apart from transmitting power from one shaft to another, gears are also used for one of the four reasons stated below:
 - 1) To reverse the direction of rotation
 - 2) To increase or decrease the speed of rotation
 - 3) To move rotational motion to a different axis
 - 4) To keep the rotation of two axes synchronised

* Types of Gears :-

1) Spur Gear :-

- It is most ^{commonly} and simplest gears designed to transmit motion between two parallel shafts.
- The axis of the two shafts i.e., the driving shaft and driven shaft are parallel to each other.
- The teeth are cut straight on the periphery of the wheel and they are parallel to the axis of the wheel.
- Spur gears are used in machine tools, gear boxes, alarm clock & watches.



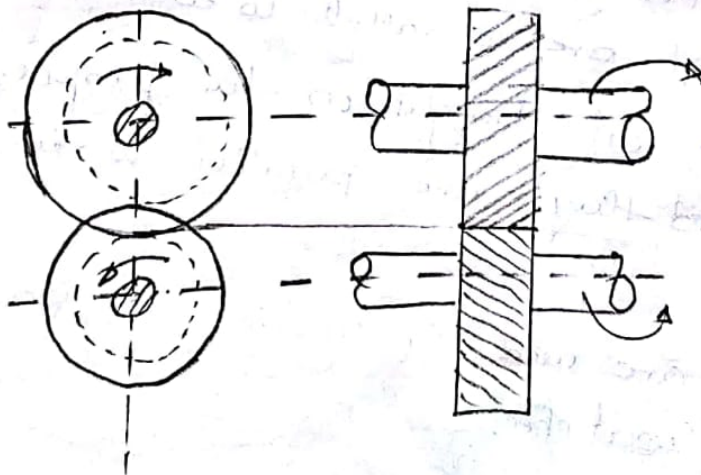
② Helical Gears :

→ Helical gears are used to transmit power between two parallel or non parallel, but non-intersecting shafts.

→ In Helical gears, the teeth are curved, and helical in shape hence the name helical gears.

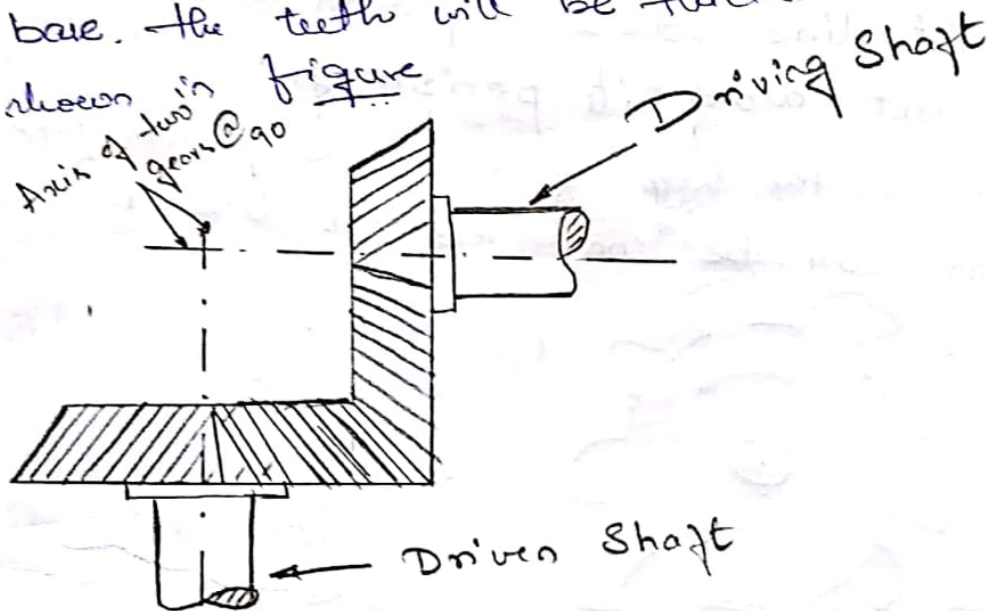
→ When two teeth on a helical gear engage, the contact starts at one end of the teeth and gradually spreads as the gears rotate, until the two teeth are in full engagement.

→ Helical gears are mostly used in automobile power transmission.



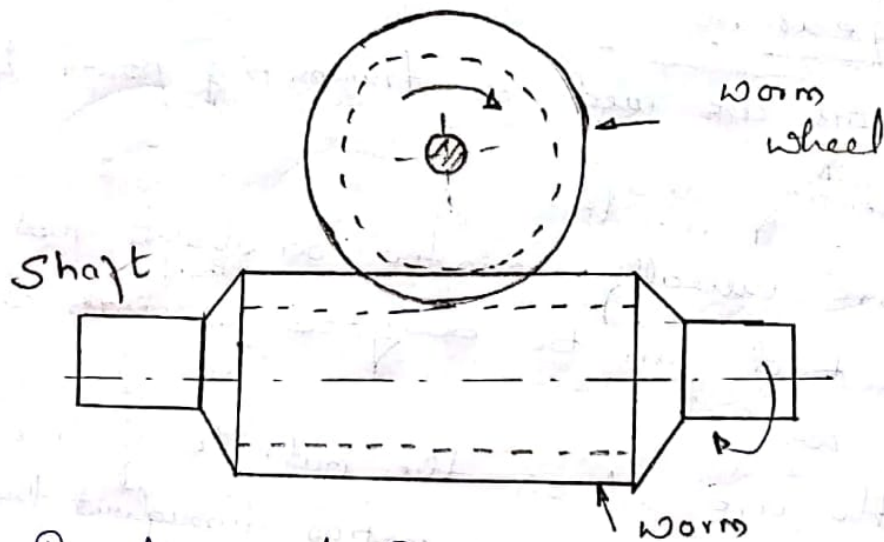
③ Bevel Gear :

- Bevel gears are used for transmitting power b/w two intersecting shafts.
- They are usually mounted on shafts that are 90° apart, but can be designed to work at other angles as well.
- The teeth are cut on the outside of the conical surface and vary in cross-section throughout their length.
- Since the diameter of the cone is greatest at its base, the teeth will be thicker at the base as shown in figure.



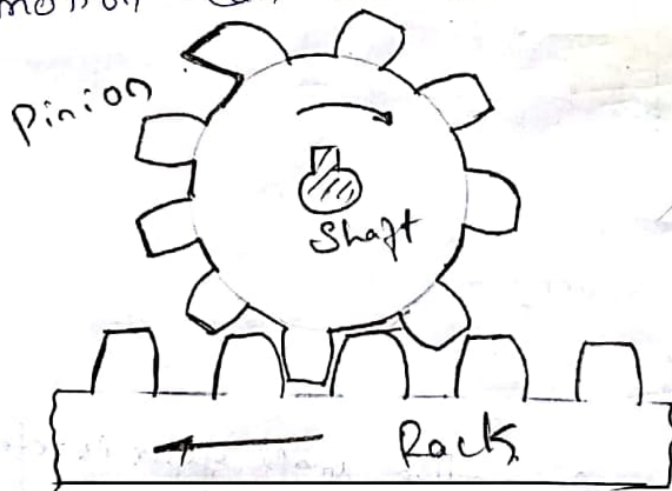
④ Worm gears :

- In worm wheel gearing, the small gear is referred to as the worm gear and the large gear as the worm wheel.
- This arrangement is used for transmission of power at high velocity ratio.
- Worm gears are used in machine tools where large speed reductions are needed.



⑤ Rack and Pinion

- A rack is a gear, having teeth cut along a straight line, while a pinion is a gear with teeth cut along its periphery.
- with the help of rack and pinion, rotary motion can be converted into linear motion.



* Advantages and Disadvantages of gear drive over belt drive.

Advantages:

- 1) For small centre distances it is effective drive to transmit power.
- 2) High power as well as low power can be transmitted.

- 3) Constant velocity ratio can be maintained
- 4) The drive is very compact
- 5) High velocity ratio can be achieved.
- 6) Operation is very simple
- 7) It is a position drive

Disadvantages:

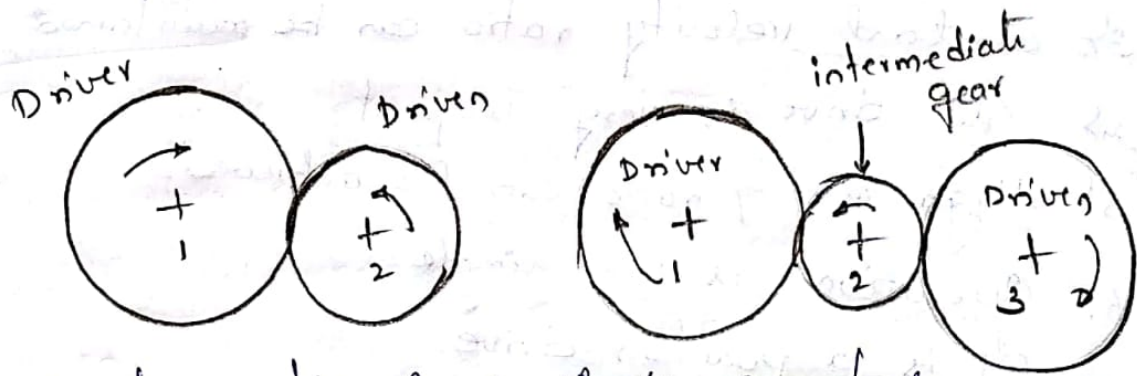
- 1) If centre distance is more it is not an economical drive.
- 2) Lubrication is essential
- 3) Initial cost is high.

* Gear Train:

- Sometimes, two or more gears are made to mesh with each other to transmit power from one shaft to other. Such an arrangement is called gear train.
- There are mainly two types of gear trains.
 - 1) Simple gear train. 2) Compound gear train.

* Simple gear train:

- The arrangement of gears in series is shown in fig (a) is a simple gear train.
- In fig (a) two gears are arranged, the direction of the driven is opposite to the direction of driver.
- In fig (b) & (c) intermediate gears are provided between the driver and driven.
- The function of the idler gears is
 - 1) To cover the space between the driver gears
 - 2) To obtain the desired direction of driven.



Velocity ratio of a simple gear train

Let N_1 , N_2 & N_3 be their speeds and Z_1 , Z_2 & Z_3 be the number of teeth on them

When gears 1 & 2 are engaged with each other

$$VR = \frac{N_2}{N_1} = \frac{Z_1}{Z_2}$$

When gears 2 & 3 are engaged

$$VR = \frac{N_3}{N_2} = \frac{Z_2}{Z_3}$$

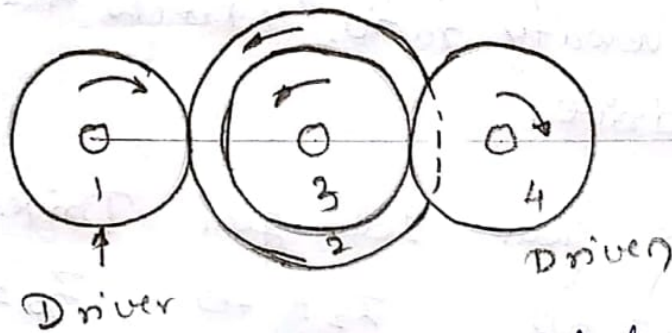
∴ Now velocity ratio b/w gear 1 & 3

$$\frac{N_1}{N_3} = \frac{N_1}{N_2} \times \frac{N_2}{N_3} = \frac{Z_2}{Z_1} \times \frac{Z_3}{Z_2}$$

$$\boxed{\frac{N_1}{N_3} = \frac{Z_3}{Z_1}}$$

* Compound gear train

In this type of gear train, each shaft carries two or more gears except the first & last shaft. All the gears revolve about a fixed axis.



Velocity ratio is calculated as follows

Gear 1 meshes with gear 2

$$\therefore VR = \frac{N_1}{N_2} = \frac{Z_2}{Z_1}$$

Gear 3 meshes with gear 4

$$VR = \frac{N_3}{N_4} = \frac{Z_4}{Z_3}$$

V.R for gear ① & ④

$$\frac{N_1}{N_4} = \frac{N_1}{N_2} \times \frac{N_3}{N_4} = \frac{N_1}{N_3} \times \frac{N_3}{N_4} \quad (N_3 = N_2)$$

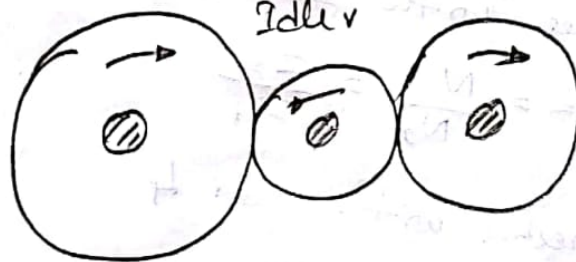
$$\boxed{\frac{N_1}{N_4} = \frac{Z_2}{Z_1} \times \frac{Z_4}{Z_3}}$$

Problems

① A simple gear train consists of 3 gears. The number of teeth on the driving gear is 60, on the intermediate (idler) gear 40, & on the driven gear 80. If the driving gear rotates at 1200 rpm, find the speed of the driven gear, and also the velocity ratio. sketch the arrangement of the gear drive.

Soln.:-

Driving gear	Idler gear	Driven gear
$Z_1 = 60$	$Z_2 = 40$	$Z_3 = 80$
$N_1 = 1200 \text{ rpm}$		$N_3 = ?$



$$\frac{N_3}{N_1} = \frac{Z_1}{Z_3}$$

$$\frac{N_3}{1200} = \frac{60}{80}$$

$$\boxed{N_3 = 900 \text{ rpm}}$$

$$\text{Velocity ratio} = \frac{N_3}{N_1} = \frac{900}{1200} = \frac{3}{4}$$

② A gear train consists of 4 gears A, B, C & D of 20, 25, 50 and 75 teeth respectively. A meshes with C & B is a compound gear with C. B meshes with D. If A has a speed of 300 rpm. what is the speed of D. Sketch the gear train.

Given data

Gear A

$$Z_A = 20$$

$$N_A = 300 \text{ rpm}$$

Gear B

$$Z_B = 25$$

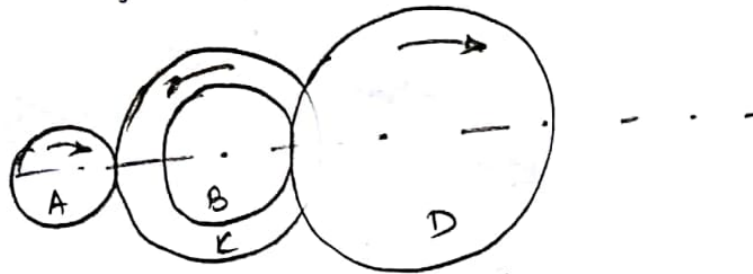
Gear C

$$Z_C = 50$$

Gear D

$$Z_D = 75$$

$$N_D = ?$$



$$\text{Velocity ratio} = \frac{\text{Speed of last gear}}{\text{Speed of first gear}}$$

$$= \frac{N_D}{N_A} = \frac{Z_A}{Z_C} \times \frac{Z_B}{Z_D}$$

$$\frac{N_D}{300} = \frac{20}{50} \times \frac{25}{75}$$

$$\boxed{N_D = 40 \text{ rpm}}$$